

GNU nano 8.7.1

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    else if (pid == 0) {
        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());

        printf("Executing command: %s\n", command);

        execlp(command, command, NULL);

        printf("Command execution failed!\n");
    }

    else {
        printf("Parent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

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badari_narayana@Badari-Narayana:~$ mkdir OS_LAB
mkdir: OS_LAB: File exists
badari_narayana@Badari-Narayana:~$ cd OS_LAB
badari_narayana@Badari-Narayana:~/OS_LAB$ nano task1.c
badari_narayana@Badari-Narayana:~/OS_LAB$ gcc task1.c -o task1
badari_narayana@Badari-Narayana:~/OS_LAB$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [371 kB]
Hit:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [432 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [89.4 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [32.7 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [321 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [105 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [61.2 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [62.0 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [161 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [331 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [63.0 kB]
Fetched 2348 kB in 6s (391 kB/s)
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
badari_narayana@Badari-Narayana:~/OS_LAB$ sudo apt install build-essential
Installing:
  build-essential

Installing dependencies:
  bzip2                libalgorithm-merge-perl
  dpkg-dev              libcrypt-dev
  fakeroot              libdpkg-perl
  g++                   libfakeroot
  g++-15                libfile-fcntllock-perl
  g++-15-x86-64-linux-gnu libstdc++-15-dev
  g++-x86-64-linux-gnu  lto-disabled-list
  libalgorithm-diff-perl make
  libalgorithm-diff-xs-perl
```

```
badari_narayana@Badari-Narayana:~/OS_LAB$ ./task1
Enter a Linux command: ls
Parent Process
Parent PID : 958
Child PID : 959
Child Process
Child PID : 959
Executing command: ls
task1 task1.c
Child process completed.
badari_narayana@Badari-Narayana:~/OS_LAB$ uname -a
Linux Badari-Narayana 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
badari_narayana@Badari-Narayana:~/OS_LAB$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      16
On-line CPU(s) list:        0-15
Vendor ID:                   GenuineIntel
Model name:                  13th Gen Intel(R) Core(TM) i7-13620H
CPU family:                  6
Model:                       186
Thread(s) per core:         2
Core(s) per socket:         8
Socket(s):                   1
Stepping:                    2
BogoMIPS:                    5836.80
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
                             se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                             _tsc rep_good nopl xtopology tsc_reliable nonst
                             op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                             sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                             opcnt tsc_deadline_timer aes xsave avx f16c rdr
                             and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                             brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                             d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi
                             2 erms invpcid rdseed adx smap clflushopt clwb
                             sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                             vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
                             vdiri movdir64b fsrm md_clear serialize ibt flu
                             sh_lld arch_capabilities
```

Virtualization features:

Virtualization: VT-x
Hypervisor vendor: Microsoft
Virtualization type: full

Caches (sum of all):

L1d: 384 KiB (8 instances)
L1i: 256 KiB (8 instances)
L2: 10 MiB (8 instances)
L3: 24 MiB (1 instance)

NUMA:

NUMA node(s): 1
NUMA node0 CPU(s): 0-15

Vulnerabilities:

Gather data sampling: Not affected
Ghostwrite: Not affected
Indirect target selection: Not affected
Itlb multihit: Not affected
L1tf: Not affected
Mds: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Old microcode: Not affected
Reg file data sampling: Mitigation; Clear Register File
Retbleed: Mitigation; Enhanced IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
Srbds: Not affected
Tsa: Not affected
Tsx async abort: Not affected
Vmscape: Not affected

badari_narayana@Badari-Narayana:~/OS_LAB\$ ps

PID	TTY	TIME	CMD
327	pts/0	00:00:00	bash
1003	pts/0	00:00:00	ps

badari_narayana@Badari-Narayana:~/OS_LAB\$ ps -ef

UID	PID	PPID	C	STIME	TTY	TIME	CMD
root	1	0	0	08:57	?	00:00:00	/sbin/init
root	2	1	0	08:57	hvc0	00:00:00	/init
root	7	2	0	08:57	hvc0	00:00:00	plan9 --control-socket 7
root	47	1	0	08:57	?	00:00:00	/usr/lib/systemd/systemd
systemd+	79	1	0	08:58	?	00:00:00	/usr/lib/systemd/systemd

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root      47      1  0 08:57 ?        00:00:00 /usr/lib/systemd/systemd
systemd+  79      1  0 08:58 ?        00:00:00 /usr/lib/systemd/systemd
root      85      1  0 08:58 ?        00:00:00 /usr/lib/systemd/systemd
root     105      1  0 08:58 ?        00:00:00 /bin/sh /usr/lib/systemd
root     106      1  0 08:58 ?        00:00:00 /usr/sbin/cron -f -P
message+ 107      1  0 08:58 ?        00:00:00 @dbus-daemon --system --
root     111      1  0 08:58 ?        00:00:00 /usr/bin/python3 /usr/bi
root     115      1  0 08:58 ?        00:00:00 /usr/lib/systemd/systemd
root     123      1  0 08:58 ?        00:00:00 /usr/libexec/wsl-pro-ser
syslog   182      1  0 08:58 ?        00:00:00 /usr/sbin/rsyslogd -n -i
root     185      1  0 08:58 tty1     00:00:00 /usr/sbin/agetty --nores
_chrony  215     105  0 08:58 ?        00:00:00 /usr/sbin/chronyd -n -F
_chrony  225     215  0 08:58 ?        00:00:00 /usr/sbin/chronyd -n -F
root     238      1  0 08:58 ?        00:00:00 /usr/bin/python3 /usr/sh
root     325      2  0 08:58 ?        00:00:00 /init
root     326     325  0 08:58 ?        00:00:00 /init
badari_+ 327     326  0 08:58 pts/0    00:00:00 -bash
root     328      2  0 08:58 ?        00:00:00 login -- badari_narayana
badari_+ 370      1  0 08:58 ?        00:00:00 /usr/lib/systemd/systemd
badari_+ 372     370  0 08:58 ?        00:00:00 (sd-pam)
badari_+ 400     328  0 08:58 pts/1    00:00:00 -bash
root     585      1  0 09:03 ?        00:00:00 /usr/libexec/packagekitd
polkitd  590      1  0 09:03 ?        00:00:00 /usr/lib/polkit-1/polkit
badari_+ 1013     327  0 09:07 pts/0    00:00:00 ps -ef
```

badari_narayana@Badari-Narayana:~/OS_LAB\$ top

```
top - 09:08:43 up 10 min,  1 user,  load average: 0.01, 0.04, 0.01
Tasks:  27 total,   1 running,  26 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  7771.6 total,  7158.1 free,   591.9 used,   178.8 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7179.8 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24272	15452	11624	S	0.0	0.2	0:00.58	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.00	init-systemd(Ub
7	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
47	root	19	-1	50400	16828	15540	S	0.0	0.2	0:00.13	systemd-journal
79	systemd+	20	0	22416	14504	12008	S	0.0	0.2	0:00.10	systemd-resolve
85	root	20	0	35424	12300	9160	S	0.0	0.2	0:00.42	systemd-udev
105	root	20	0	2888	1948	1820	S	0.0	0.0	0:00.03	chronyd-starter
106	root	20	0	4472	3160	2908	S	0.0	0.0	0:00.00	cron
107	message+	20	0	8812	5356	4636	S	0.0	0.1	0:00.05	dbus-daemon
111	root	20	0	44096	29932	14792	S	0.0	0.4	0:00.18	networkd-dispat
115	root	20	0	18436	9444	8204	S	0.0	0.1	0:00.07	systemd-logind
123	root	20	0	1792300	12784	10152	S	0.0	0.2	0:00.10	wsl-pro-service
182	syslog	20	0	220548	5716	4432	S	0.0	0.1	0:00.06	rsyslogd
185	root	20	0	5492	2748	2564	S	0.0	0.0	0:00.00	agetty
215	_chrony	20	0	20424	6408	5632	S	0.0	0.1	0:00.04	chronyd
225	_chrony	20	0	12096	2404	1760	S	0.0	0.0	0:00.00	chronyd
238	root	20	0	123460	32644	18032	S	0.0	0.4	0:00.17	unattended-upgr
325	root	20	0	3184	1108	980	S	0.0	0.0	0:00.00	SessionLeader
326	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.29	Relay(327)
327	badari_+	20	0	6268	5528	3856	S	0.0	0.1	0:00.09	bash
328	root	20	0	8652	5592	4744	S	0.0	0.1	0:00.01	login
370	badari_+	20	0	22340	13540	10924	S	0.0	0.2	0:00.12	systemd
372	badari_+	20	0	22916	4088	2088	S	0.0	0.1	0:00.00	(sd-pam)
400	badari_+	20	0	6136	5416	3864	S	0.0	0.1	0:00.02	bash
585	root	20	0	340156	22236	19168	S	0.0	0.3	0:00.04	packagekitd
590	polkitd	20	0	306444	8476	7660	S	0.0	0.1	0:00.06	polkitd
1025	badari_+	20	0	8328	6092	3888	R	0.0	0.1	0:00.06	top